CSM
Fall 2025

EECS 16B
Worksheet 9

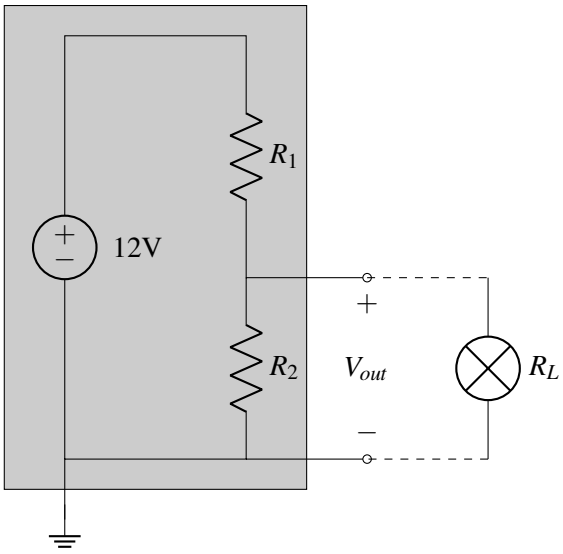
1. Op Amps as Buffers

Learning Goal: This problem helps understand the operating principle of an op-amp buffer and how it helps with loading.

Relevant Notes: goes different op-amp circuit topology and corresponding derivations.

Now we will revisit a problem that you might have seen before, with our new knowledge of op-amps. We have access to a circuit inside a 'black box' as shown below, with two terminal coming out of it.

(a) We need a voltage of 6V power a light bulb with resistance R_L. Design R_1 and R_2 inside the black box so that the voltage across R_2 is exactly equal to this required voltage **when the bulb is not connected**; i.e. $V_{R_2} = V_{out} = 6\text{V}$.



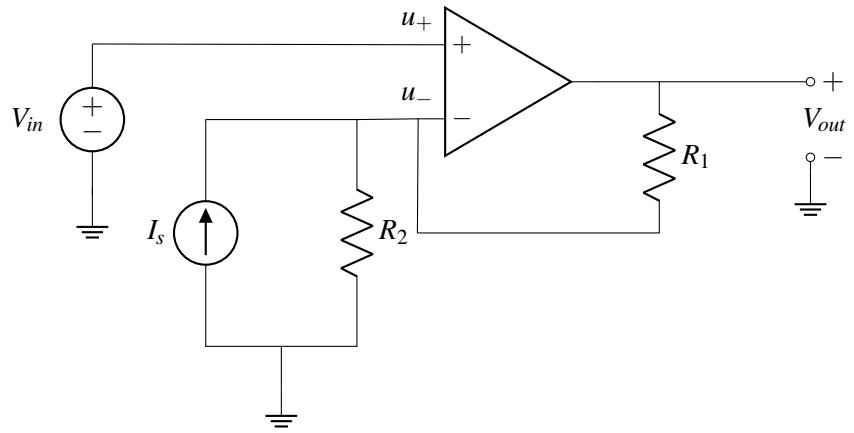
(b) Now let us connect the bulb R_L across R_2. What is the voltage across R_1, R_2 and R_L when the bulb is connected when $R_L = R_2$? Use the values of R_1 and R_2 from the last part. Will the light bulb turn on? What happens if $R_L = 2R_2$?

(c) Using your knowledge of op-amps, how could you resolve this issue of V_{out} changing based on the value of R_L? Think about how you might use an op-amp buffer.

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2. Introduction to Op-Amp Analysis

Learning Goal: The goal of this question is to build an understanding of negative feedback and op amps.



(a)

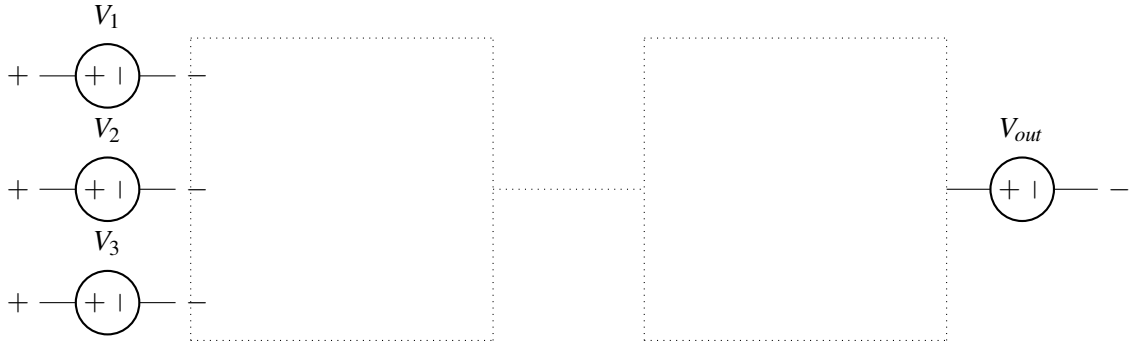
(b) Is this circuit in negative feedback?

(c) Solve for V_{out} as a function of V_{in}, I_s, R_1, and R_2.

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3. Inverting Multiple Voltage Sources

Suppose we have some speaker system in which we receive three different DAC voltages. In order to produce our desired speaker voltage, we want to average these three DAC voltages, and reverse the voltage sign. This process can be broken down in two steps, as shown below



(a) Draw a circuit for the left box that can be used on its own to average the three voltages. You have access to unlimited resistors, but they are all 1k ohm and 2k ohm

(b) Now, draw a circuit the left box that can be used to invert the sign of our signal. Suppose you have access to one op amp and one resistor, which can have any value. Specify the value of your chosen resistor

(c) Demonstrate that the resulting circuit is in negative feedback.

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